

AP CSP | Lesson 07: Data Quality & Cleaning

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: D

D describes the meaning and unit of the recorded values. A, B and C are values in individual records, so they are primary data in this table.

Q2. Answer: B

B: editing metadata leaves the primary number unchanged. A assumes an automatic conversion. C confuses relabeling with conversion: 30 minutes is 0.5 hours. D incorrectly claims the measurement is deleted.

Q3. Answer: C

C preserves the intended Bus category. A changes the mode; B invents a numeric value; D discards a recoverable response instead of normalizing an equivalent label.

Q4. Answer: A

A follows the rule and preserves an audit trail. B treats unknown as measured zero; C invents a value from another response; D destroys the source evidence. The exclusion applies to this time analysis.

Q5. Answer: B and D

B and D. B retains one response without double-counting; D flags uncertainty without assuming an error. A loses the response entirely. C gives no evidence for exclusion. One mark only for B and D together.

Q6. Answer: C

C uses the shared identifier to match the same responses. A can pair different IDs. B invents matches from similarity. D ignores different quantities and units.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: (a) "One-way time in minutes" defines quantity and unit. (b) The fictional, voluntary, single-school context identifies source and scope. (c) No: it stays 30. (d) $30 \div 60 = 0.5$ hours.

1 mark: Identifies a field definition/unit and explains how it helps interpret values.

1 mark: Identifies collection/source metadata and explains its scope or limitation.

1 mark: States that changing only metadata leaves the number 30 unchanged.

1 mark: Gives 0.5 hours, equivalent to 30 minutes.

Q8. 4 marks

Answer: (a) 02: 25 min, 4.0 km; 03: 12 min, 0.8 km. (b) Different row orders would pair different responses. (c) Check compatible ID meanings, units, field definitions or collection periods.

1 mark: Matches ID 02 to 25 minutes and 4.0 km.

1 mark: Matches ID 03 to 12 minutes and 0.8 km.

1 mark: Explains that positional matching would pair different IDs.

1 mark: Gives a relevant check of ID meaning, units, definitions or period.

Q9. 4 marks

Answer: (a) 09 is unknown; 10 is invalid negative elapsed time. (b) Keep both raw rows; exclude both from this analysis and log each reason. (c) Neither a measured zero nor an intended +5 is established.

1 mark: Identifies 09 as missing and 10 as invalid negative elapsed time.

1 mark: Preserves both raw rows and excludes them from this analysis.

1 mark: Logs missing time for 09 and negative time for 10.

1 mark: Explains that both replacements invent unsupported measurements.

Q10. 4 marks

Answer: (a) $16 - 1 - 1 - 1 = 13$. (b) Normalize “ bus ” to Bus; keep the row. (c) Retain and flag 120: no error is established.

1 mark: Shows three exclusions and calculates 13 retained rows.

1 mark: Normalizes ID 15 to Bus, preserving the meaning.

1 mark: States that normalization does not reduce the row count.

1 mark: Retains ID 14 because unusual alone does not establish an error.

Q11. 4 marks

Answer: (a) Conflicting values do not identify the correct time. (b) Check original submissions or correction records. (c) Keep raw evidence, flag uncertainty and document the handling rule, such as temporary exclusion. (d) Log ID/original values and action/reason.

1 mark: Explains why a conflicting repeat cannot be resolved as an identical copy.

1 mark: Gives a relevant source or collection-record check.

1 mark: Preserves raw evidence and documents an unresolved flag or justified handling rule.

1 mark: Names two useful log items: ID, values, action, reason or evidence.

Q12. 4 marks

Answer: (a) Mode and minutes; a response ID only if needed. (b) Identifying fields or linked attributes can reveal identity. (c) No: cleaning does not make the sample representative. (d) No: the same selection/source bias may remain.

1 mark: Keeps mode/time and omits unnecessary names/addresses.

1 mark: Explains a concrete identification risk from fields or linked data.

1 mark: Distinguishes clean values from a representative sample.

1 mark: Explains why more same-source responses need not remove selection bias.